

In addition to the online homework, please hand in the following problems:

Problem 1: Find an infinite series of non-zero terms whose sum is:

- (i) 1
- (ii) -3
- (iii) 0

Problem 2: Show by example that $\sum_{n=1}^{\infty} \frac{a_n}{b_n}$ may diverge even though $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge and $b_n \neq 0$ for every n .

Problem 3: If $A = \sum_{n=1}^{\infty} a_n$ converges and $a_n > 0$ for all n , can anything be said about the convergence of $\sum_{n=1}^{\infty} \frac{1}{a_n}$? Give proof for your answer.

Problem 4: If $\sum_{n=1}^{\infty} a_n$ converges and $\sum_{n=1}^{\infty} b_n$ diverges, can anything be said about the convergence of $\sum_{n=1}^{\infty} (a_n + b_n)$? Give proof for your answer.

Problem 5: Find the value of b for which $1 + e^b + e^{2b} + e^{3b} + \dots = 9$

Problem 6: For what values of r does the infinite series $1 + 2r + r^2 + 2r^3 + r^4 + 2r^5 + r^6 + \dots$ converge? Find the sum of the series when it converges.